

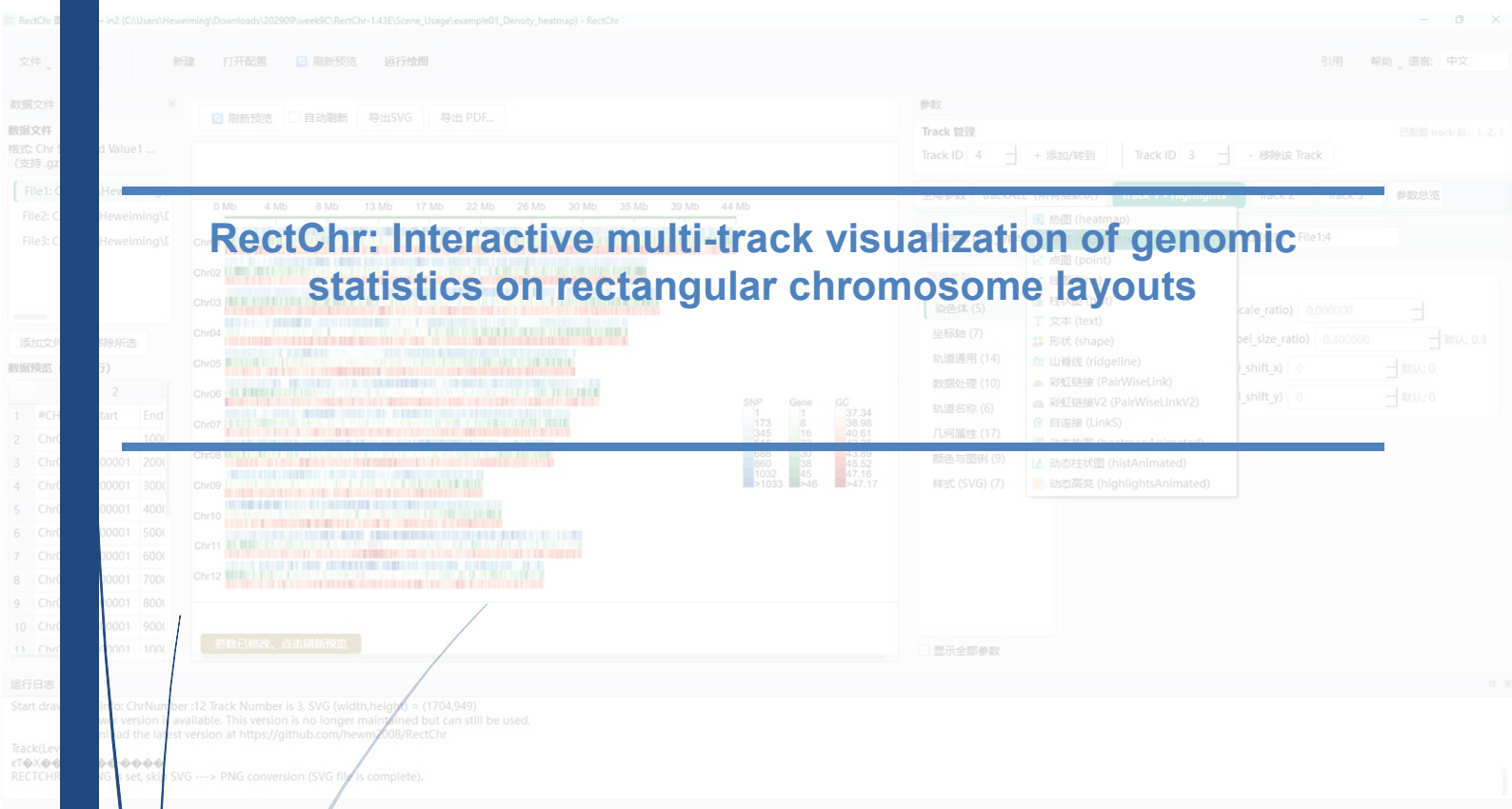
User guide



RectChr

Graphical interface

RectChr: Interactive multi-track visualization of genomic statistics on rectangular chromosome layouts



Version 1.50 User Documentation

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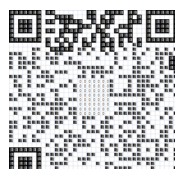


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1 Overview

The RectChr GUI is the **graphical user interface for RectChr**. It supports data management, **track configuration, drawing-parameter settings, plot preview and figure export**. This manual is designed for new users and anyone who need to adjust figures interactively, guiding them through the complete workflow-from opening a configuration file to saving the final output.

The program runs on Windows, Linux, and macOS; The GUI handles configuration and preview, while image generation is handled by *the Perl* engine. The GUI is suitable for interactive exploration, publication-quality figure, and configuration reuse. For batch processing, RectChr can be incorporated into its command-line workflow.

Main features

- ✚ Create, open, edit, and save configuration files.
- ✚ Import multiple data files and preview the first 20 rows of the selected file.
- ✚ Add, switch, and remove tracks, configure global parameters, default track parameters, and individual track parameters.
- ✚ Refresh plots manually or automatically, view run logs, and export figures in SVG, PDF, or PNG format.

Startup the program

Windows: Double click **“RectChrGUI.bat” in RectChr package**.

Linux or macOS: In the terminal, go to the directory containing the startup script, then run:

```
sh RectChrGUI.sh
```

Installing Dependencies on Windows for First Use

If you are running RectChr from the source directory on Windows, install the following dependencies first:

1. Install Perl:<https://strawberryperl.com/>. RectChr uses the Perl engine to generate figures. During installation, keep the default options selected so that Perl is added to your system PATH.
2. Install Python:<https://www.python.org/downloads/windows/>. During installation, select Add python.exe to PATH.
3. Install PySide6: Open PowerShell and run “pip install PySide6”.

After installation, double-click **RectChrGUI.bat** to start the program. On the first launch, the startup script will automatically try to install PySide6; **Perl must be installed in advance**; the script does not install it automatically

Typical workflow

1. Create a new configuration or open an existing *.conf* configuration file.
2. Add and check data files to confirm chromosome names, coordinates, and numeric columns.
3. Create the required tracks and specify the data source and plotting type for each track.
4. Adjust global parameters, track default parameters, and individual track parameters.
5. Refresh the preview, check the plot and run logs.
6. Run the plot and export the figure
7. Save the configuration for future modification and reproduction.

2 Interface Overview

The main interface includes the **top menu** and quick-operation area, the **data panel** on the left, the **plotting preview** in the center, the **parameter panel** on the right, and the **run log** and **status bar** at the bottom. The screenshot in this guide uses the English interface. Button names in the text correspond to those shown in the screenshots.

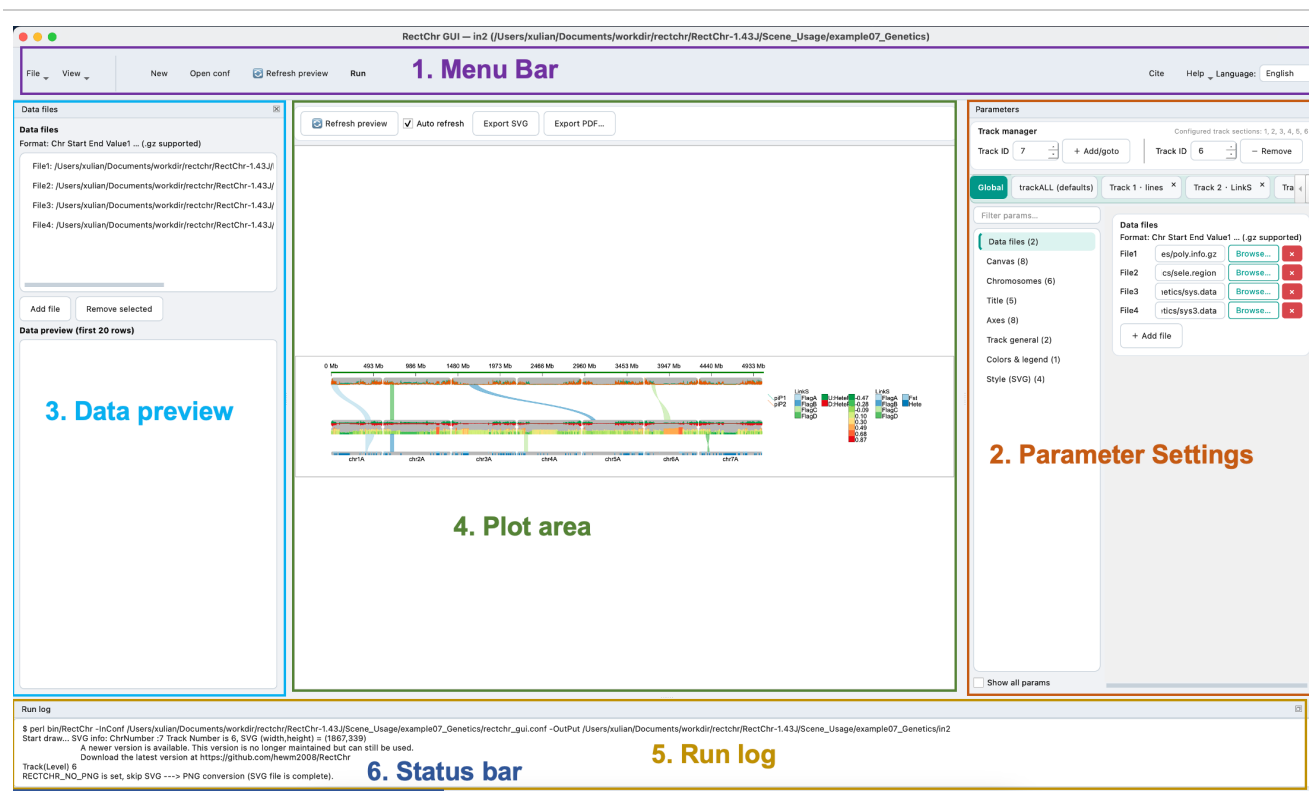


Figure 1: RectChr GUI main interface and six panels

Panels	Location	Main uses
1 Menu bar	Top	New, open, and save configurations; Refresh and run; View reference information, help, and switch languages.
2 Parameter panel	Right side	Manage Tracks; Switch parameter views; Filter, enable, and edit parameters.
3. Data panel	Left side	Manage input files and preview the first 20 lines of the selected file.
4. Plot area	Central	Display drawing results; Refresh preview, zoom to view, and export drawings.
5 Run log	Bottom	Check the runtime, warnings, and errors.
6 Status Bar	Window bottom	View the current status and related paths.

First-time users can go directly to Chapter 3, Quick Start. Data formats, plot types, parameter settings, and troubleshooting are covered in Chapters 4, 5, 6, and 9, respectively

Menus and Shortcuts



Figure 2 Configuration Management, View Settings, and Common actions

The top toolbar provides common actions, including creating a new configuration, opening an existing configuration, refreshing the preview, and running a plot. Configuration saving and figure export are available from the **File** menu.

New	Create a blank drawing task.
Open conf	Open the existing .conf configuration file.
File → Save conf	Save the current configuration.
Refresh preview	Update the preview according to the current settings.
Run	Run the current drawing task and check its execution in the Run log.
File → Export SVG / PNG / PDF	Export the corresponding graphic format.
View	Show or hide switchable panels; Revert the interface layout via Reset layout.
Cite / Help	Check for references or help.
Language	Switch interface language.

Tip: Before modifying existing tasks, it is recommended to save a copy of the configuration first; If the panel is accidentally closed or the layout changes, you can first check the View menu.

Data files and Previews

The Data files panel on the left lists the currently loaded input files. Select a file to display its first 20 rows in Data preview. This allows you to quickly verify the number of columns, delimiters, and field contents.

The screenshot shows the 'Data files' panel in the RectChr software. It lists four files with their full paths. Below the list are buttons for 'Add file' and 'Remove selected'. A 'Show first 20 lines' button is also present. Below this is a 'Data preview (first 20 rows)' table with 8 columns: #Chr, Start, End, piP1, piP2, HeteP1, HeteP2, and Fst. The table displays data for 15 rows, all starting with 'chr1A'.

	1	2	3	4	5	6	7	8
1	#Chr	Start	End	piP1	piP2	HeteP1	HeteP2	Fst
2	chr1A	1	1	0.4015	0.3748	0.4005	0.3723	0.0557
3	chr1A	100000	100000	0.3589	0.3100	0.3581	0.3080	0.0356
4	chr1A	200000	200000	0.3825	0.3065	0.3816	0.3045	0.0022
5	chr1A	300000	300000	0.8278	0.5843	0.8259	0.5805	-0.0777
6	chr1A	400000	400000	1.0157	0.7088	1.0133	0.7042	-0.0830
7	chr1A	500000	500000	1.0559	0.8855	1.0534	0.8797	-0.0124
8	chr1A	600000	600000	1.1260	0.9070	1.1234	0.9011	-0.0265
9	chr1A	700000	700000	1.0996	0.8104	1.0969	0.8051	-0.0599
10	chr1A	800000	800000	0.8143	0.6628	0.8124	0.6584	-0.0217
11	chr1A	900000	900000	1.7074	1.4718	1.7034	1.4622	-0.0227
12	chr1A	1000000	1000000	2.4757	2.1882	2.4698	2.1738	-0.0155
13	chr1A	1100000	1100000	1.5736	1.4771	1.5698	1.4674	0.0113
14	chr1A	1200000	1200000	0.7041	0.6798	0.7024	0.6753	0.0182
15	chr1A	1300000	1300000	0.0776	0.0847	0.0774	0.0842	0.0606

Figure 3: Select the data file and check the first 20 lines

Before plotting, check the following:

- ✚ The selected file exists and its path is valid.
- ✚ Chromosome names, start and end coordinates, values, and annotation columns are as expected.
- ✚ The first 20 lines are only for quick inspection. Confirm that the format is consistent throughout the complete file before generating a final plot.
- ✚ Before removing a file, check whether any track still references it.

Parameter Panel

The parameter panel on the right organizes settings by scope and function. First select the scope you want to edit, then locate and modify the appropriate parameters.

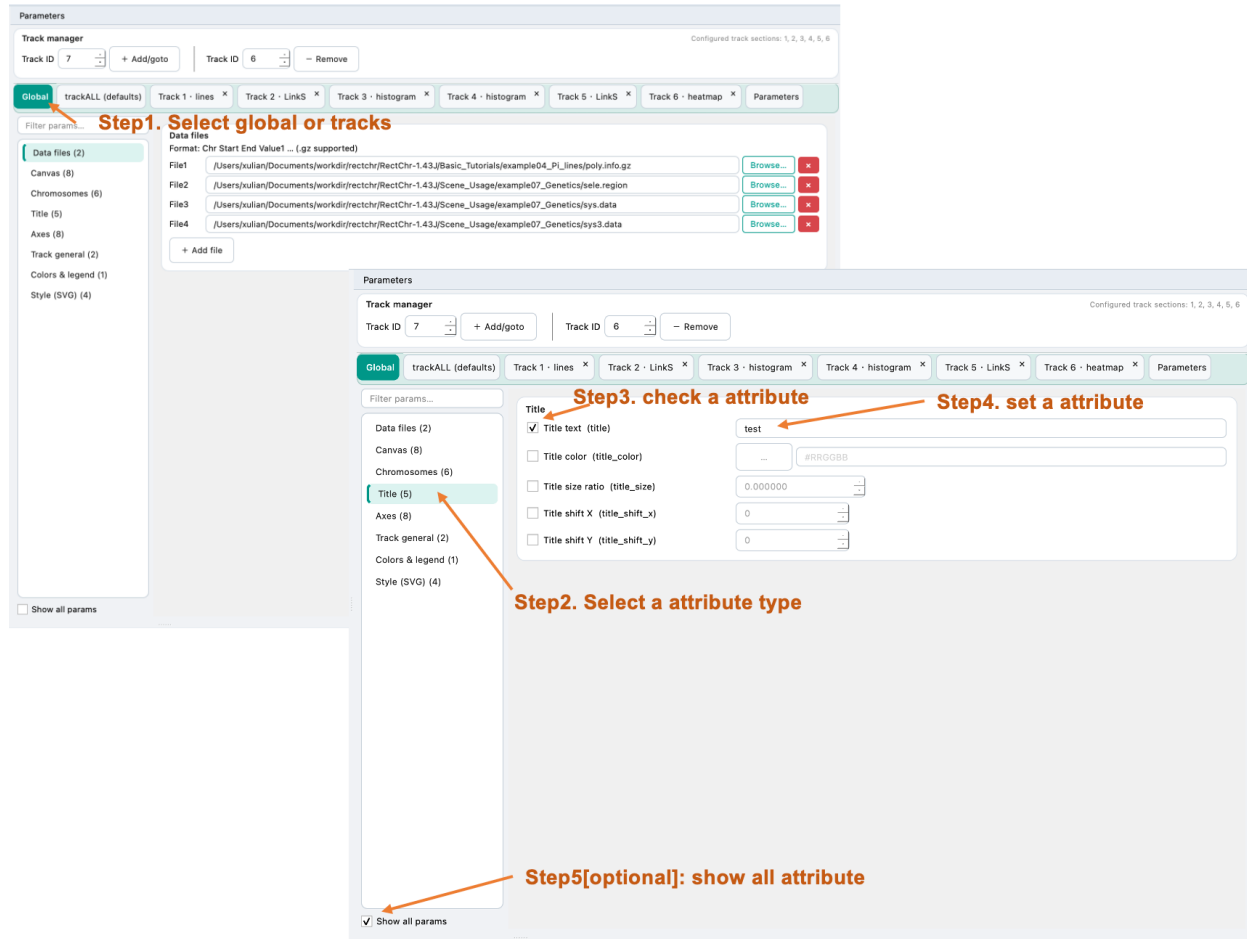


Figure 4 Select parameter range, classify parameters, and enable required parameters

1. Select the Global, trackALL (default settings), or target Track tab.
2. In the category list, select a parameter group, such as Title or Axes.
3. For a parameter with a checkbox, select the checkbox first to enable its editing control.
4. Enter or modify the value, then refresh the preview to check the effect.
5. To access additional options, select Show all params. You can also use Filter params to search for parameters by keyword.

Individual track settings and parameters overview

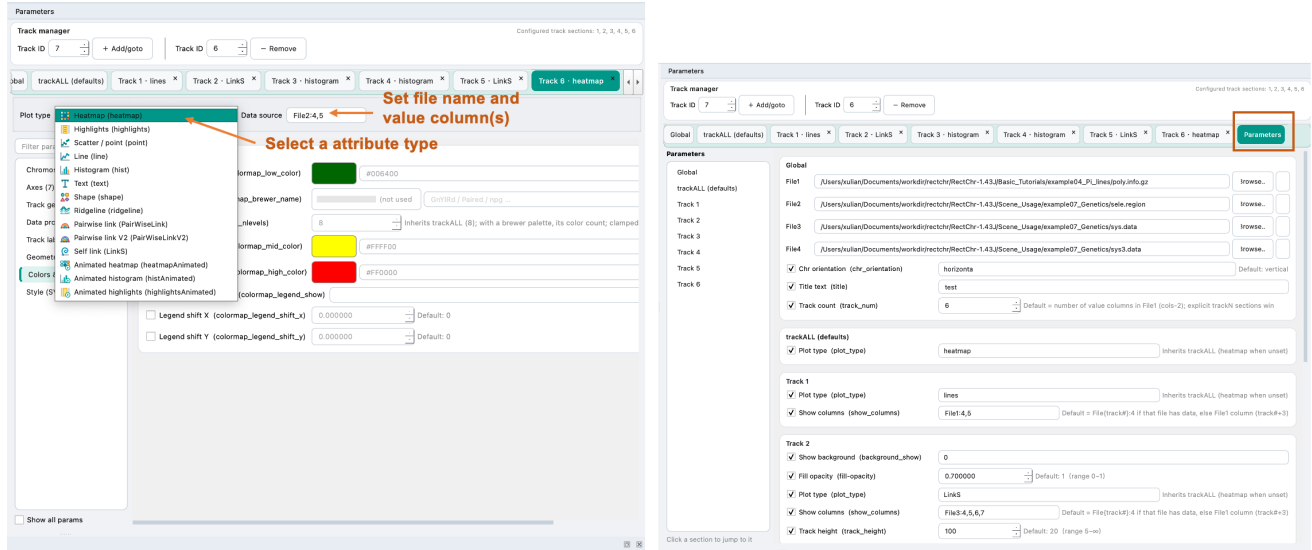


Figure 5 Sets the drawing type and data source for a single Track, and views the parameter overview

Each **Track tab** represents one plotting track. Use **Plot type** to select the plot type, then specify the input file and columns under Data source. The **Parameters tab** provides a centralized view of the current configuration for all scopes.

View	Scope of action	Typical uses
Global	The entire graphic or task	Input files, canvases, chromosome layouts, and main titles.
trackALL (defaults)	Default settings for all tracks	Set styles or axis rules shared by multiple tracks.
Track X	Specify the Track	Set the drawing type, data source, and independent style for the track.
Parameters	Current configuration overview	Centrally verify parameters for global, default, and each Track.

For example, **File2:4,5** in Figure 5 refers to columns 4 and 5 of File2. File and column numbers must match the actual input data. Required columns and their meanings vary by plot type, so do not directly reuse example column numbers for data with a different structure.

Recommended approach: Configure shared default settings in trackALL first, then override only the settings that need to differ in individual **Track tabs**.

3. Quick Start

Open an existing configuration

1. Start **RectChrGUI** and click Open conf.
2. Select an existing *.conf* file, or open the example configuration from the *Basic_Tutorials* and *Scene_Usage* directory in the RectChr package.
3. Check the path in the Data files; If input data has been moved, assign the correct files again.
4. Click **Refresh Preview** or **Run**, then inspect the plot and the Run log.
5. Adjust parameters as needed, save the configuration, and **export** the figure.

Create a new plotting task

Start with a blank configuration and follow these three steps to complete the first figure:

Add data

1. Click **New** to create a blank task.

2. In **Data files**, click **Add file** and select an input file.
3. Select the added file and check its column count, delimiters, and initial rows in **Data Preview**.

Configuring the first Track

1. Select **Track 1** in the Track Manager; If it does not exist, enter its ID in **Add/goto** and click **Add/goto**.
2. In **Global** → **Data files**, confirm that **File1** points to the intended input file..
3. Switch to **Track 1**, select a **Plot type**, and specify the required files and columns in Data source.
4. Set the necessary chromosomal ranges and layout parameters; Adjust color, title, and axis as needed.
5. Click **Refresh Preview** to check the plot orientation, track position, and labels.

Generating and save Results

1. Click **Run** or **Refresh preview** and confirm that the run log contains no errors that prevent plotting.
2. Export the figure as SVG, PDF, or PNG from the **File** menu.
3. Select **File** → **Save conf** to save the current configuration.

For your first plot, retain most default settings and configure only the required data, plot type, and chromosome parameters. After confirming that the figure can be generated successfully, adjust the appearance gradually.

Minimum heatmap configuration

For a first plot, keep most default settings and specify only the required data, plot type, and chromosome parameters. Once plotting succeeds, adjust the appearance gradually.

```
SetParaFor=global
File1 = ./data.gz
```

4. Data and track management

Basic input requirements

The interface generally expects chromosome interval data separated by spaces or tabs. A typical structure is:

- ```
Chr Start End Value1 ...
```
- ✚ **Chr**: Chromosome name. It must match the chromosome name used in the chromosome-length or karyotype configuration.
  - ✚ **Start** and **End**: Interval coordinates. Both must be numeric, and Start must not exceed End.
  - ✚ **Value1** and **subsequent columns**: Numeric values or annotation fields. Their meanings depend on the selected plot type.

The interface supports for .gz files, so compressed data files can be selected directly.

Different plot types may require different numbers of columns and field definitions. Before plotting, consult the relevant examples to confirm the required input format and data-column settings.

### Adding and removing files

File operations are available from both the **left-side data panel** and the **parameter panel**. After adding a file, verify its path, format, delimiters, and column structure before assigning it to a track.

| Operation     | Method                                                                                                   | Key points to check                                                    |
|---------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Add</b>    | Click Add file on the left, or use Browse in the parameter panel.                                        | The file appears in the list, and its first 20 lines can be previewed. |
| <b>Append</b> | Click Add file in the parameter panel to add another input.                                              | Configure the corresponding data source for the tracks that need it.   |
| <b>Remove</b> | After selecting the file, click Remove selected, or use the red delete button in the parameters section. | Check whether any track still references the file.                     |

|                |                                  |                                                                |
|----------------|----------------------------------|----------------------------------------------------------------|
| <b>Replace</b> | Use Browse to select a new path. | Refresh the preview and check the new file's column structure. |
|----------------|----------------------------------|----------------------------------------------------------------|

## Adding and removing Tracks

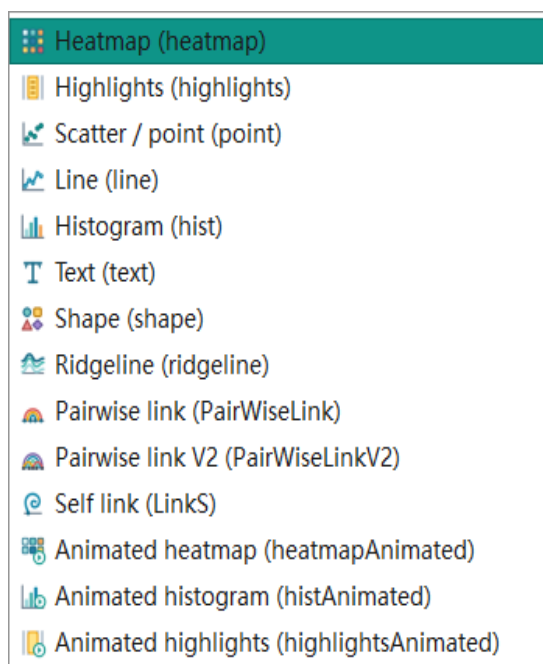
A Track is a layer of data or annotation drawn along the chromosome layout. Tracks can represent different samples, data types, or statistics. Track IDs identify the objects being edited and generally affect the arrangement of tracks; always confirm the actual order in the preview.

1. In Track manager, enter the target Track ID beside Add/goto, then click **Add/goto** to add or select it.
2. Configure the plot type, data source, and style in the corresponding **Track tab**.
3. To remove a track, confirm its ID in beside Remove, then click **Remove**.

*Note: Removing a track may also remove its dedicated settings. Save a copy of the configuration before deletion.*

## 5 Plot Types (plot\_type)

RectChr provides **14 plot types**. Select a type from the *plot\_type* drop-down list in a **Track tab**; the form updates to show the relevant parameters.



*plot\_type* accepts several alternative spellings, such as line/lines and hist/histogram, and normalizes them automatically when the configuration is loaded or saved.

### Heatmap

Maps regional statistics to colored cells. Commonly used for genomic density, Fst, and depth.

**main parameters:** colormap\_low\_color, colormap\_mid\_color, colormap\_high\_color, colormap\_nlevels, Ymax, Ymin

```
| plot_type = heatmap
```

### Highlights

Colors interval backgrounds to mark selected regions, QTL intervals, centromeres, and similar features

**main parameters:** background\_color, track\_bg\_height\_ratio

```
| plot_type = highlights
```

---

## Point/Scatter

Displays individual points; commonly used for GWAS Manhattan plots

**main parameters:** cutoff\_y, cutoff\_color, log\_p, track\_point\_size

```
plot_type = point
```

## Line

Uses lines or curves to show trends in continuous values

**main parameters:** track\_height, Ymax, Ymin

```
plot_type = line
```

## Histogram

Displays bars. Specifying two columns in show\_columns produces opposing upper and lower histograms.

**main parameters:** show\_columns, track\_height, yaxis\_tick\_show

```
plot_type = hist
show_columns = File1:6,7
```

## Text

Displays labels, such as gene or marker names, at the corresponding regions.

**main parameters:** track\_text\_size, track\_text\_angle, track\_text\_anchor

```
plot_type = text
show_columns = File2:5
```

## Shape

Draws geometric shapes at regions, such as circles, squares, and triangles, using shape codes 0–13.

**main parameters:** track\_geom\_shape, track\_geom\_shape\_size

```
plot_type = shape
track_geom_shape = 3
```

## Ridgeline

Displays stacked profiles, such as QTL signals or distributions.

**main parameters:** track\_height

```
plot_type = ridgeline
```

## Rainbow Links (PairWiseLink)

Draws rainbow links between two genomes or intervals to show synteny or variation.

**Main parameters:** link\_direction, link\_linestyle, link\_uniform\_height

```
plot_type = PairWiseLink
```

## PairWiseLinkV2

An extended version of rainbow links, suited to large-scale synteny visualization.

**Main parameters:** link\_direction, link\_linestyle, link\_uniform\_height

```
plot_type = PairWiseLinkV2
```

## Self-Linking (LinkS)

Connects regions within a chromosome.

**Main parameters:** link\_direction, link\_linestyle

```
plot_type = LinkS
```

## Animated Heatmap (heatmapAnimated)

Creates an animated SVG heatmap. Open the exported SVG in a browser to play the animation.

**Main parameters:** colormap\_\*, Ymax, Ymin

```
plot_type = heatmapAnimated
```

### Animated Histogram (histAnimated)

Creates an animated SVG histogram. Open it in a browser to play the animation.

**Main parameters:** track\_height, show\_columns

```
plot_type = histAnimated
```

Creates animated SVG highlights. Open the exported SVG in a browser to play the animation.

**Main parameters:** background\_color, track\_bg\_height\_ratio

```
plot_type = highlightsAnimated
```

## 6 Parameter configuration

### Parameter Scope and activation

**Global**, **trackALL** (defaults), and **Track X** are used for global settings, track defaults, and individual track settings, respectively; The **Parameters** tab provides a centrally view configurations. Before modifying parameters, first confirm the current tab and track number to avoid adjusting to the wrong object.

Before editing, verify the current tab and track number to avoid changing the wrong scope.

For parameters with checkboxes, select the checkbox before editing the value. Use Filter params to find a parameter, and select Show all params to access options hidden in the simplified view.

### Common parameter classifications

| Interface classification | Main uses                                               | Usage recommendations                                                                   |
|--------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Data files               | Specify input files and related data.                   | First, check the data preview and file number.                                          |
| Canvas                   | Set canvas size, margins, and overall layout.           | Adjusted according to the final layout proportions.                                     |
| Chromosomes              | Set the range, sequence, and appearance of chromosomes. | Keep the input consistent with the name in the configuration.                           |
| Title                    | Set the title and related text styles.                  | The title should be concise, and the font size should match the final output size.      |
| Axes                     | Set scales, labels, numeric ranges, and shaft styles.   | Maintain comparable scales when comparing multiple tracks.                              |
| Track general            | Set general track layout rules.                         | Set the default value first, then adjust the single track.                              |
| Colors & legend          | Set up color mapping and legend.                        | Color schemes should be easy to distinguish, with clear meanings for units and legends. |
| Style (SVG)              | Set the relevant styles for vector output.              | Before post-editing, check text, line width, and style.                                 |

### Recommended adjustment order

1. Data files, data columns, and plot types.
2. Chromosome range and order.
3. Track height, spacing, and axis range.
4. Colors, lines, and legends.
5. Titles, fonts, and tags.
6. Canvas size and output format.

First confirm that the data mapping and plot structure are correct. Then refine the visual style, and finally check readability at the intended output size.

## 7 Preview and Plotting

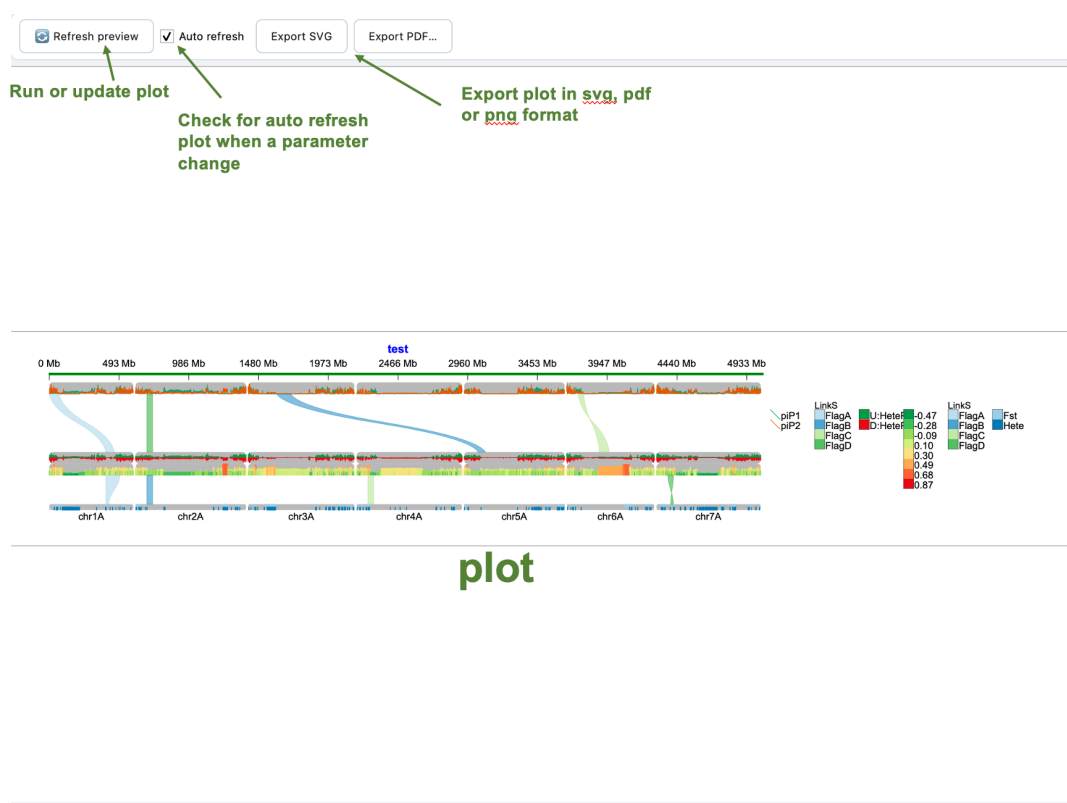


Figure 6: Preview refresh, auto-refresh, and graphic export entry

### Manual and automatic refresh

Click **Refresh Preview** to update the current plot. When Auto refresh is enabled, the preview updates automatically after parameter changes.

**For large input files, many tracks, or a series of related adjustments, disable automatic refresh and refresh manually after completing a group of changes.** Use the mouse wheel to zoom the preview.

### Preview checklist

Before exporting, verify that:

- Target chromosomes are fully displayed in the expected order, direction, and range.
- Each track uses the intended file and data columns, without unexpected empty areas or duplicate content.
- Axis ranges, units, and directions are correct, and extreme values do not obscure the main data distribution.
- Titles, labels, and legends are readable and do not overlap or extend beyond plot boundaries.
- Colors are distinguishable; check grayscale appearance if necessary.

### Checking the run log

After selecting **Run** or refreshing the preview, review the Run log at the bottom of the window for run information, output paths, warnings, and errors.

If the preview is empty, keep the complete log and first check the file path, input format, and required parameters.

## 8 Save Configurations and Export Figures

### Save and reopen a configuration

1. After confirming that the plot runs correctly, select File → Save conf.
2. Save a new version before making major changes, for example, project\_v01.conf and project\_v02.conf.
3. After reopening a configuration with Open conf, check all input-data paths.
4. When moving a project between computers, copy the configuration and data together while preserving the project directory structure. Reassign file paths if necessary.

Configuration files preserve plotting settings. Exported figures are used for viewing, sharing, or later editing. Keep both the configuration file and the original input data.

### Choosing the output format

The characteristics of the three output formats and key points to check after export are as follows:

| Format | Features and applicable scenarios                                                                           | Inspection after export                                            |
|--------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| SVG    | Vector format, suitable for continuing to edit text and graphics in software like Illustrator and Inkscape. | Font replacement, line width, transparency, and the edited layout. |
| PDF    | Suitable for sharing, reviewing, printing, and archiving.                                                   | Page size, font display or embedding, and transparency effects.    |
| PNG    | Bitmap format suitable for quick sharing, presentations, or scenarios requiring bitmaps.                    | Pixel size and clarity; May be distorted when zoomed in.           |

### Export figures

1. Refresh or run the plot and confirm that it meets your requirements.
2. In the plot area, click Export SVG or Export PDF. Alternatively, select Export SVG, Export PDF, or Export PNG from the File menu.
3. Choose a save location and complete the export.
4. Open the exported file and inspect text, legends, lines, and overall completeness at the intended final size

**Recommendation for submission:** Keep the original data, configuration file, editable SVG, and final submission files. If a bitmap is required, verify that its resolution meets journal requirements. Do not retain only screenshots of the interface.

## 9. Troubleshooting

If a problem occurs, check the Run log first, then verify the file path, data structure, and required parameters. Change and refresh one type of setting at a time to help identify the cause.

### Common Phenomena and Treatment

| Phenomenon                  | Possible reasons                                                                           | Handling methods                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Blank data preview          | Invalid path, empty file, abnormal compressed file, or delimiter not meeting requirements. | Reselect the file, check its size, and use the text tool to view the first few lines.                 |
| No figure in the plot area  | Missing required parameters, Track not bound to data, or task execution failure.           | Check input, drawing type, and logs; Retain necessary settings to temporarily restore default styles. |
| Missing chromosomes         | Inconsistent names or filtered by range parameters.                                        | Standardize chr1 and 1 names; Check chromosome range and sequence settings.                           |
| Empty Track                 | Incorrect column number, missing values, or coordinates out of range.                      | Check file numbers, column definitions, numeric types, and Start and End coordinates.                 |
| Parameters cannot be edited | The corresponding checkbox has not yet been checked.                                       | First, check the box on the left side of the parameter; Enable Show all params if necessary.          |
| Overlapping labels          | The canvas is too small, the font too large, or the track spacing is insufficient.         | Increase the canvas or spacing, shorten the label, or appropriately reduce the font size.             |

|                                                      |                                                                            |                                                                                                        |
|------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Slow automatic refresh</b>                        | Large data volumes, multiple tracks, or continuous triggers for redrawing. | Turn off Auto refresh, complete a set of changes, then manually refresh.                               |
| <b>Export differs from preview</b>                   | Fonts, transparency, or size perform differently in different renderers.   | Check output dimensions and styles, use common fonts, and check the final export file.                 |
| <b>Files missing after reopening a configuration</b> | Data paths change or projects are moved.                                   | Respecifying file paths; Copy data together when used across computers and retain directory structure. |

## Other tips

- ✚ “No Perl found” on Windows: Extract the portable version of Strawberry Perl to `gui_runtime/perl/`, or install Perl and ensure that it is available through the system PATH.
- ✚ Opening an older configuration: Legacy parameter names are mapped automatically, for example, `cutoff_value` to `cutoff_y` and `strokewidth` to `stroke-width`. Manual changes are not required.
- ✚ Animated plots: `heatmapAnimated`, `histAnimated`, and `highlightsAnimated` appear as static frames in the preview. Open the exported SVG in a web browser to play the animation.

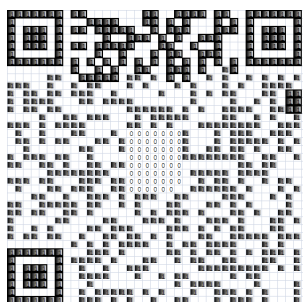
## 10. Contact Information

Use Help to access the Chinese or English manual and About, and use Cite for citation information. When reporting a problem, provide the configuration, a minimal dataset that reproduces the issue, the complete run log, and your operating system and software version

### Contact details

- email: ✉ [hewm2008@gmail.com](mailto:hewm2008@gmail.com) / [hewm2008@qq.com](mailto:hewm2008@qq.com)
- home: <https://github.com/hewm2008/RectChr>
- QQ group: 125293663

WeChat tipping QQ group joining: **125293663 WeChat public**



群名称: Reseqtools (itools)  
群 号: 125293663

